

Digital Logic Design Report

Problem Analysis and Arduino Implementation

Bheemraj Paidipelli

M.Sc Artificial Intelligence & Data Science

June 8, 2026

Table of Contents

Contents

1	Abstract	3
2	Objective	3
3	Problem Statement	3
4	Circuit Analysis	4
5	Truth Table Verification	5
6	Hardware Implementation	5
6.1	Components Required	5
6.2	Pin Assignment	5
6.3	Circuit Connections	6
6.4	Working Principle	6
7	Arduino Program	7
8	Assembly Program	7
9	Results	7
10	Optimisation Analysis	7
10.1	Gate Count Comparison	7
10.2	Advantages of Simplification	7
11	Conclusion	8

1 Abstract

This report presents the analysis and implementation of a digital logic circuit using Boolean algebra and Arduino UNO. The given circuit contains three input variables U , V , and W . The corresponding Boolean expression is derived from the logic diagram and implemented in software using Arduino programming. The system reads the input states through push buttons and computes the output according to the Boolean expression

$$I = (U + V')(U + W)(V + W')$$

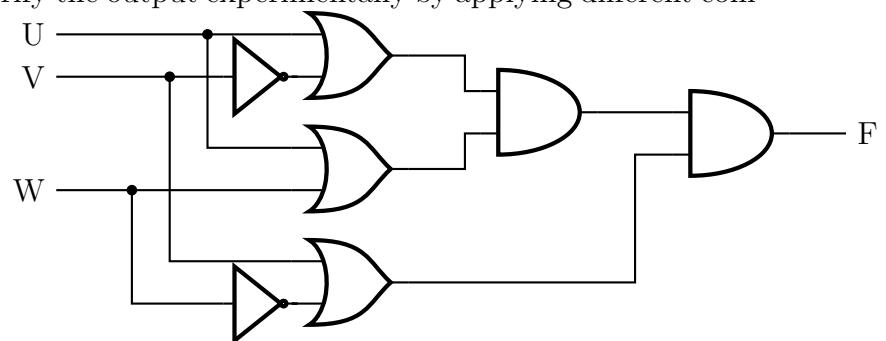
The output is displayed using an LED connected to the Arduino board. Experimental testing verifies the correctness of the logic function for different combinations of input values.

2 Objective

1. To derive the Boolean expression from the given logic circuit.
2. To implement the logic function using Arduino UNO.
3. To verify the output for different input combinations.
4. To understand the relationship between Boolean algebra and digital hardware implementation.
5. To analyze the operation of logic gates using practical experimentation.

3 Problem Statement

Write the Boolean expression represented by the given logic circuit and implement the expression using Arduino UNO. Verify the output experimentally by applying different combinations of inputs U , V , and W .



4 Circuit Analysis

The given logic circuit consists of three OR gate outputs that are combined using AND operations to generate the final output.

The outputs of the three OR gates are:

$$X_1 = U + \overline{V}$$

$$X_2 = U + W$$

$$X_3 = V + \overline{W}$$

The final output I is obtained by ANDing these three intermediate outputs:

$$I = X_1 \cdot X_2 \cdot X_3$$

Substituting the values of

$$X_1 \cdot X_2 \cdot X_3$$

:

$$I = (U + \overline{V})(U + W)(V + \overline{W})$$

Thus, the Boolean expression representing the given logic circuit is:

$$I = (U)(V + \overline{W})$$

This Boolean expression is directly implemented in the Arduino program using logical OR (||), logical AND (&&), and logical NOT (!) operators to determine the output for different input combinations.

Verified Result

$$I = (U)(V + \overline{W})$$

The output of the circuit depends on the values of the inputs U, V, and W according to the simplified Boolean expression above. Experimental verification using Arduino UNO showed that the output matched the expected results for all input combinations. This confirms that the simplified expression accurately represents the behaviour of the original logic circuit and can be used for its implementation.

5 Truth Table Verification

U	V	W	Output I
0	0	0	0 ✖
0	0	1	0 ✖
0	1	0	0 ✖
0	1	1	0 ✖
1	0	0	1 ✔
1	0	1	0 ✖
1	1	0	1 ✔
1	1	1	1 ✔

Table 1: *Truth Table for the Simplified Boolean Expression $I = U(V + \overline{W})$*

6 Hardware Implementation

6.1 Components Required

Component	Quantity
Arduino UNO	1
Push Buttons	3
LED	1
220 Ω Resistor	1
Breadboard	1
Jumper Wires	As Required

Table 2: Components Required

6.2 Pin Assignment

Arduino Pin	Function
D2	Input U
D3	Input V
D4	Input W
D13	Output I (LED)

Table 3: Arduino UNO Pin Assignment

6.3 Circuit Connections

1. Connect one terminal of the push button representing U to pin D2 and the other terminal to GND.
2. Connect one terminal of the push button representing V to pin D3 and the other terminal to GND.
3. Connect one terminal of the push button representing W to pin D4 and the other terminal to GND.
4. Connect the anode of the LED to pin D13 through a $220\ \Omega$ resistor.
5. Connect the cathode of the LED to GND.
6. Enable the internal pull-up resistors using the `INPUT_PULLUP` mode in the Arduino program.

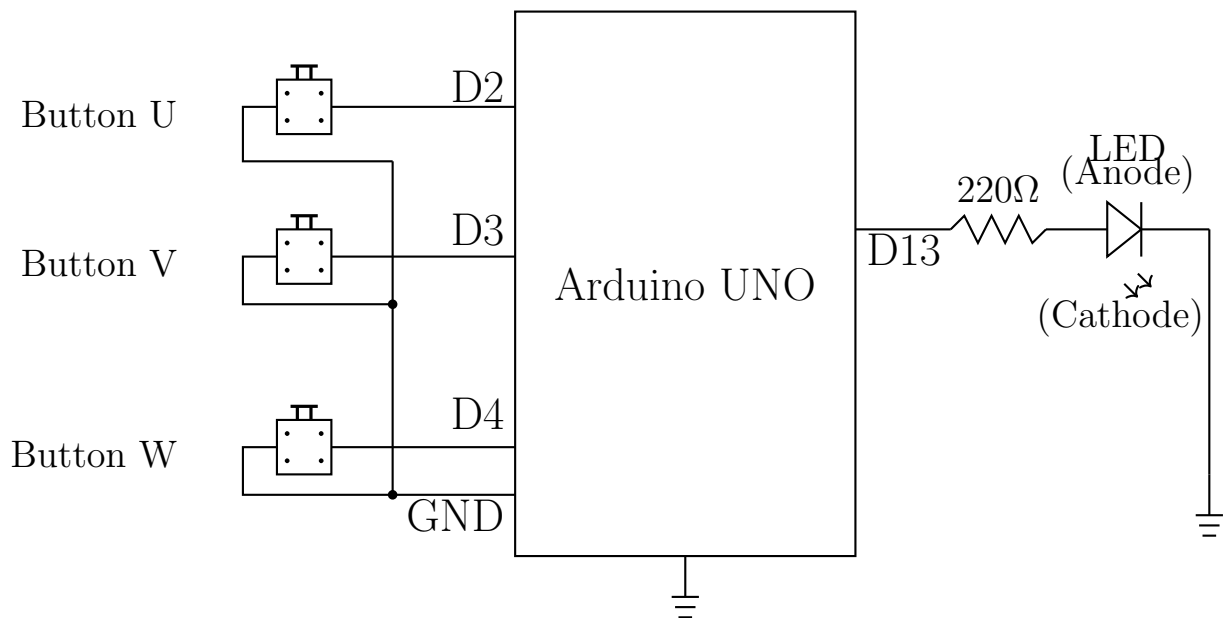


Figure 1: Hardware Connection Diagram

6.4 Working Principle

The Arduino UNO continuously reads the states of the three input switches connected to pins D2, D3, and D4. Since the inputs are configured using `INPUT_PULLUP`, pressing a switch produces a logic HIGH after software inversion.

The Arduino evaluates the Boolean expression

$$I = U(V + \overline{W})$$

and sends the result to the LED connected to pin D13.

When the expression evaluates to logic HIGH, the LED turns ON. When the expression evaluates to logic LOW, the LED remains OFF.

The values of U , V , W , and the corresponding output I are also displayed on the Serial Monitor, allowing real-time verification of the logic circuit operation.

7 Arduino Program

Full source code on GitHub:

<https://github.com/Bheemrajpaidipelli/IITH-Research-May/blob/main/gateproblem.ino>

8 Assembly Program

Full source code on GitHub:

<https://github.com/Bheemrajpaidipelli/IITH-Research-May/blob/main/gateproblem.asm>

9 Results

Experimental Findings

The Arduino UNO successfully evaluated the simplified Boolean expression for all possible combinations of the input variables U , V , and W . In each case, the output LED connected to pin D13 produced the expected logic state, experimentally confirming:

$$I = U(V + \overline{W})$$

Both the C++ (Arduino) implementation and the AVR Assembly implementation produced identical results, demonstrating consistency across both levels of abstraction.

10 Optimisation Analysis

10.1 Gate Count Comparison

Metric	Original Expression	Simplified Expression
OR Operations	3	1
NOT Operations	2	1
AND Operations	2	1
Logic Levels	3	2
Implementation	Complex Logic	Reduced Logic

Table 4: Original Expression vs. Simplified Expression

10.2 Advantages of Simplification

- **Reduced logic complexity** — the original Boolean expression is simplified to a more compact form.
- **Fewer logical operations** — the number of OR, NOT, and AND operations is reduced.
- **Lower propagation delay** — fewer logic levels result in faster output generation.

- **Simpler implementation** — the simplified expression is easier to implement in both hardware and software.
- **Improved readability** — the simplified equation is easier to understand, analyse, and debug.
- **Efficient resource utilisation** — fewer logic operations require less computational effort during execution.

11 Conclusion

The given logic circuit was successfully analysed and the corresponding Boolean expression was derived from the circuit diagram. Using Boolean algebra, the expression was simplified to

$$I = U(V + \overline{W})$$

The simplified expression was implemented using an Arduino UNO and verified experimentally using push-button inputs and an LED output. The observed output matched the expected truth table values for all possible input combinations, confirming the correctness of both the circuit analysis and the simplification process.

The AVR Assembly implementation also produced identical results, demonstrating consistency between high-level and low-level implementations. This project highlights the importance of Boolean simplification in reducing logic complexity, improving implementation efficiency, and facilitating the design and verification of digital electronic systems.

— End of Report —